Regulatory data sheet (RDS) - Pharmaceutical market (human and veterinary medicines)

MONOSTEOL (Code: 3079)

Last update: 06 December 2023

1. DEFINITION

DEFINITION
Propylene glycol monopalmitostearate EP

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Propylene Glycol Stearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	F76354LMGR
ADDITIONAL INFORMATION ON THE DENOMINATION	PROPYLENE GLYCOL MONOSTEARATE [NF], PROPYLENE GLYCOL STEARATE [INCI] PROPYLENE GLYCOL MONOPALMITOSTEARATE PROPYLENE GLYCOL PALMITOSTEARATE
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Propylene glycol and Stearic acid 50. No solvent or catalyst used during the manufacturing process.

COMPOSITION
Well defined multi-constituent substance composed of Propylene glycol mono- and di- esters of Palmitostearic acids (C16 and C18), the monoester fraction being predominant (Specification > 50%)
Full Labeling (Key letters system) **$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$**

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COMPOSITION	
Propylene Glycol Stearate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. DRUG MASTER FILE

A Drug Master File (DMF) is a submission to the Food and Drug Administration (FDA) that may be used to provide confidential detailed information about facilities, processes, or articles used in the manufacturing, processing, packaging, and storing of one or more human drugs. The submission of a DMF is not required by law or FDA regulation. A DMF is submitted solely at the discretion of the holder.
(Source: US FDA)

In general, the excipient DMF is not used for compendial excipients but more for novel excipients (e.g., new excipients, mixtures, coprocessed materials, and biotech excipients).
FDA usually does not review DMFs for compendial excipients unless there is a specific reason to do so (e.g., new route of administration or application for existing excipients).
(Source: IPEC Americas)

We have no DMF submitted to FDA for MONOSTEOL because it is a compendial excipient.

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopoeia (Ph.Eur.) Current edition	Propylene Glycol Monopalmitostearate
USP/ NF Current edition	Not conform to USP-NF Propylene Glycol Monostearate

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6. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	**Ingredient Name**	**Use**	**Restrictions**
Approved	Propylene glycol monostearate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicine	Only for use in topical medicines for dermal application. Listed Medicines
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved chemical name	Propylene Glycol Stearate	Role Non-medicinal: Purposes: Skin-Conditioning Agent Route Of administration: Topical
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

7. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSE					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute irritation (JORF)	Dermal	Rabbit	MONOSTEOL™ 20% in tween 60 + water	Slightly irritant Not classified Cutaneous irritation index: 1.13	Gattefossé data, GAT- 78185, 1978
Acute irritation (JORF)	Ocular	Rabbit	MONOSTEOL™ 20% in tween 60 + water	Slightly irritant Not classified Ocular irritation index (acute): 10.3 Ocular irritation index (48h): 0.83	Gattefossé data, GAT- 78185, 1978

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	**Reference**	**Maximum "as used" concentration**	**Conclusions**
Propylene Glycol Stearate and Propylene Glycol Stearate SE	JACT 2(5): 101-124, 1983	Up to 25%	On the basis of the available information presented in this report, the panel concludes that Propylene Glycol Stearate and Propylene Glycol Stearate SE are safe as cosmetic ingredients in the present practices of concentration and use.

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8. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI	**CAS #**	**EC #**	**Chemical name**
Propylene Glycol Stearate	91031-35-5 (or) 91672-07-0 (or) 1323-39-3	292-936-3 (or) 294-012-5 (or) 215-354-3	Fatty acids, C16-18, esters with propylene glycol (or) Octadecanoic acid, mixed esters with palmitic acid and propylene glycol (or) Octadecanoic acid, monoester with 1,2-propanediol

EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list.
EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5.
Substances without EC numbers were assigned "list numbers" in order to facilitate REACh submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.

NATIONAL INVENTORIES					
INCI	**U.S.A. (TSCA)**	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**
Propylene Glycol Stearate	Not listed (or) Not listed (or) Listed	Not listed (or) Not listed (or) Listed R-ICL listed under 91031-35-5	Not listed (or) Not listed (or) Listed	Not listed (or) Not listed (or) Listed	Fatty acids, C16-18, esters with propylene glycol: EINECS, NZIoC, REACH, VNECI / Fatty acids, (C=16-18), esters with propylene glycol: AREC (or) Octadecanoic acid, mixed esters with palmitic acid and propylene glycol: EINECS, REACH, VNECI (or) Octadecanoic acid, monoester with 1,2-propanediol: AIIC, DSL, ENCS, IECSC, INSQ, NZIoC, PICCS, REACH, TSCA / stearic acid, monoester with propane-1,2-diol: EINECS, REACH

U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation.
"Naturally occurring substances" are not subject to Premanufacture Notice reporting.

Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer.
Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.

Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities.
Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration.
Chemicals not listed on the AIIC must be assessed before importation.

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REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	**Status per substance**
Propylene Glycol Stearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

9. HISTORY OF USE

9.1. Pharmaceutical applications

9.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	**Route**	**Dosage form**	**Maximum Potency per unit dose**	**Maximum Daily Exposure (MDE)**
PROPYLENE GLYCOL MONOPALMITOSTEARATE (UNII : F76354LMGR)	TOPICAL	CREAM	9.3%w/w	
	TOPICAL	LOTION	4.69%w/w	
	TOPICAL	OINTMENT	8%w/w	
	TOPICAL	OINTMENT, AUGMENTED	2%w/w	
	VAGINAL	CREAM	7%w/w	
	VAGINAL	SUPPOSITORY	135mg	
PROPYLENE GLYCOL MONOSTEARATE (UNII: MZM1I680W0)	TOPICAL	CREAM	9.3%w/w	
	TOPICAL	LOTION	3%w/w	
	TOPICAL	OINTMENT	8%w/w	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

9.1.2. Other references

COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	**Category (concentration in %)**	**regulatory status**
Propylene glycol monostearate	Emulsion base	/
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portugese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., *Pharmaceutical Technology, Drug Delivery*, 2001, p14-24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

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9.2. Food/Nutraceutical/Dietary supplement applications

Propylene glycol esters of Fatty acids have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

Country	Reference	Registration number
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 477 Propane-1,2-diol esters of fatty acids
E.U.	EFA (European Food additives)	E 477 Propane-1,2-diol esters of fatty acids
U.S.A.	DSHEA (Dietary Supplement Health and Education Act) "Grandfathered ingredient" authorized for use in Dietary supplements	Esters of propylene glycol
U.S.A.	FCC (Food Chemical Codex)	Propylene glycol mono- and diesters
U.S.A.	USFA US Food additives according to 21 CFR	21CFR §172.856 Propylene glycol mono-and diesters of Fats and Fatty acids
U.S.A.	GRAS US Food additives according to 21 CFR	Propylene glycol monostearate
JAPAN	JSFA (Japanese Standards of Food Additives)	Propylene glycol esters of Fatty acids

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

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